

Real Analysis I — Homework 2

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Consider $\mathbb{R}^n$ and $\mathcal{M} = \mathcal{M}(\mathbb{R}^n)$. If not specified, n is arbitrary; otherwise $n = 1$.

1. Suppose $E \notin \mathcal{M}$ and $\mu(A) = 0$. Is $E \setminus A \in \mathcal{M}$? Prove your claim.
2. Show that $E \in \mathcal{M}$ iff for any $A \subset E$ and $B \subset E^c$, there is $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$.
3. Show that $E \in \mathcal{M}$ iff there exists G_δ -set H such that $E \subset H$ and $\mu^*(H \setminus E) = 0$.
4. Suppose $A \in \mathcal{M}$, and B is an arbitrary set (not necessarily in $\mathcal{M}$). Show that $\mu^*(A \cap B) + \mu^*(A \cup B) = \mu(A) + \mu^*(B)$.
5. Suppose $\mu^*(A \triangle B) = \mu^*(B \triangle C) = 0$. Show that $\mu^*(A \triangle C) = 0$.
6. Suppose A and B are bounded. Show that $|\mu^*(A) - \mu^*(B)| \leq \mu^*(A \triangle B)$.
7. Suppose $A \subset [0, 1]$ is measurable and $\mu(A) = 1$. Show that for any measurable $B \subset [0, 1]$, there is $\mu(A \cap B) = \mu(B)$.
8. Does there exist a nonempty open set G such that $\mu(G) = 0$? Prove your claim.
9. Suppose G is open and E has measure zero. Prove that $\overline{G} = \overline{G \setminus E}$.
10. Does there exist a closed proper subset $F \subsetneq [a, b]$ such that $\mu(F) = b - a$? Prove your claim.